photocatalyst-dft: where the project stands

A walkthrough of the Claude Science work, with the concepts behind each step

Prepared for Lindejiade

24 October 2026 — based on the repository state on this date

Status in one paragraph. The molecular half of the project is built and mostly validated. Sixteen organic photocatalysts are catalogued, cross-checked against PubChem, embedded in 3D and given B3LYP/def2-SVP frontier orbitals in implicit water. Six molecules have full DFT geometry optimisations. The three most recent (eosin Y^{2-}, riboflavin, rhodamine B^{+}) showed that the MMFF force-field geometries are *not* close to DFT for flexible or charged dyes (heavy-atom RMSD 0.17–0.40 Å, versus $\leq$0.03 Å for rigid molecules). A new step-5 script was written to test whether this matters for the orbital energies, but it cannot run yet, because step 3 has not been rerun on the new geometries. TDDFT and all solid-state work (Week 2) have not started.

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1 Project map

The pipeline is five scripts. Each one reads the previous step's files and writes its own, so any step can be rerun on its own.

Step	Script	What it does → main output
1	`01_represent.py`	SMILES → checks → multi-conformer 3D (ETKDGv3 + MMFF94s) → `molecules/xyz/*.xyz`, 2D grid, 3D HTML viewer
2	`02_optimize.py`	DFT geometry optimisation (PySCF + geomeTRIC) → `molecules/xyz_opt/*.xyz`, `geomopt_summary.csv`
3	`03_frontier.py`	DFT single point on the best available geometry → HOMO/LUMO for all 16, `frontier_orbitals.csv`, energy-level diagram
4	`04_orbitals.py`	Orbital cube files + fragment localisation (Mulliken) → `orbital_localization.csv`, isosurface figure
5	`05_geometry_check.py`	**New.** Does the geometry (MMFF vs DFT) change the orbital energies? → `geometry_sensitivity.csv` (*not yet produced*)
—	`utils/common.py`	Shared code: xyz read/write, building the PySCF molecule, the DFT settings, constants, plotting style

The level of theory is the same everywhere: **B3LYP/def2-SVP, density fitting, ddCOSMO water ($\varepsilon = 78.36$), restricted closed shell, no dispersion correction.**

2 Step 1 — Catalogue and molecular representation

2.1 The catalogue

`data/organic_photocatalysts.csv` has one row per molecule and 20 columns. The ones the code depends on are `slug` (the file-name ID used by every later step), `smiles`, `formula`, `charge` and `inchikey`. The rest is literature metadata: family, role, λ_{max}, the lowest absorption band and redox notes.

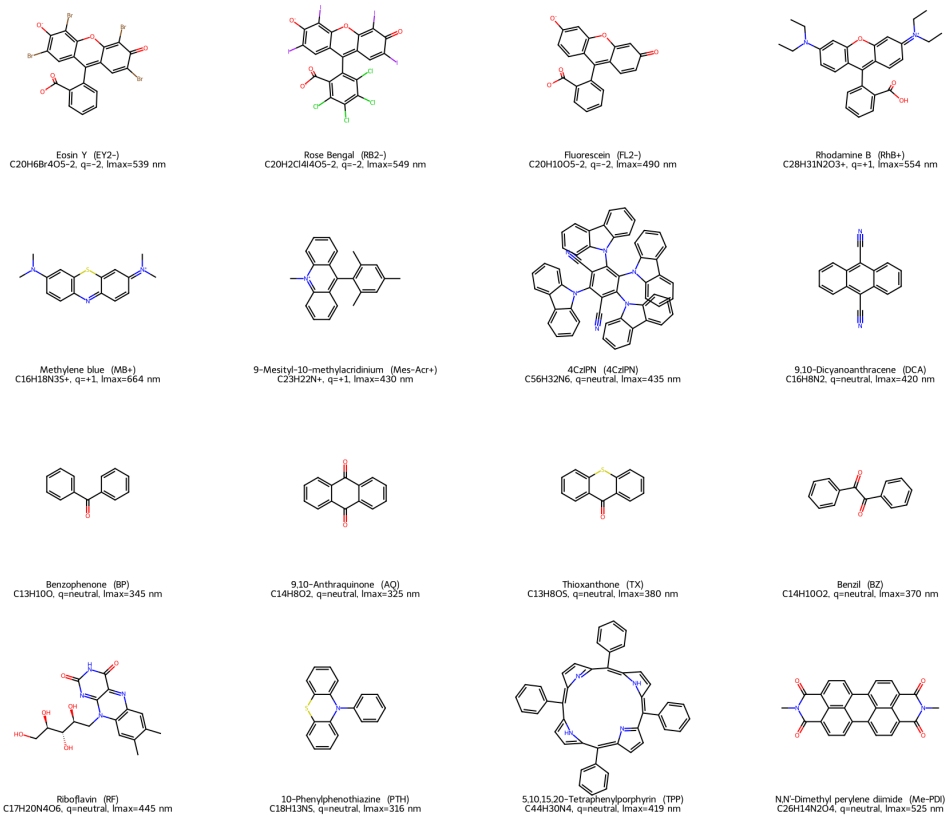


Figure 1: The 16 molecules as encoded. Charges are part of the identity: EY^{2-}, RB^{2-} and FL^{2-} are dianions; MB^{+}, RhB^{+} and Mes-Acr^{+} are cations.

PubChem cross-check. Each name was resolved against PubChem. 9 of 16 stored species match a PubChem record exactly. For the other 7, `pubchem_parent_cid` points to a *related* record, and `pubchem_cid_relation` explains how they relate:

- salts (EY, RB, RhB, MB, Mes-Acr): PubChem stores the salt; we keep only the active ion;
- fluorescein: PubChem's main record is the neutral lactone; we use the quinoid dianion;
- Me-PDI: no PubChem record; built from the parent PTCDA (CID 67191).

Concept to master

Compute the species that does the chemistry. In water at working pH, eosin Y is a dianion, and its colour and redox potentials belong to that dianion. The InChIKey records this in its last letter: neutral eosin Y ends in `-N`, the dianion in `-L` (two protons removed). The first 14 characters, which encode the skeleton, are identical. That is why a salt or ion can be traced back to its parent record in PubChem.

2.2 Checks before any 3D work

For each row the script parses the SMILES (RDKit returns `None` on a syntax error; the molecule is then skipped). It then computes formula, total charge and InChIKey from the parsed structure and compares each with the stored value. Mismatches go into a `problems` list instead of stopping the run, so all errors are reported at once.

Caveat / open issue

The formula and charge checks catch real typos. For example, dropping one `(Br)` from the eosin Y SMILES still gives a valid SMILES, but the formula becomes $C_{20}H_7Br_3O_5{}^{2-}$. The InChIKey check, however, can only catch *later edits*, because the stored key was itself generated from the same SMILES. The independent check is the PubChem comparison above.

2.3 Multi-conformer 3D embedding

1. `AddHs`: implicit hydrogens become real atoms, so they get coordinates (eosin Y: 29 → 35 atoms).
2. **ETKDGv3** distance-geometry embedding with a fixed random seed (`0xF00D`). It generates 8 conformers, or 30 if the molecule has more than 3 rotatable bonds. Conformers within 0.5 Å RMSD of each other are pruned as duplicates.
3. Every conformer is minimised with **MMFF94s**, and the lowest-energy one is written to xyz with `charge=... mult=1` in the comment line. The xyz format has no charge field; `read_xyz` recovers it with the regular expression `charge=(-?\d+)`.

Molecule	Rotatable bonds	Conformers tried	Kept after pruning
anthraquinone	0	8	1
eosin Y^{2-}	2	8	2 (mirror images, identical energy)
riboflavin	9	30	18

Concept to master

The number of surviving conformers measures flexibility. A rigid molecule collapses to one structure however many times you embed it. Riboflavin's ribityl chain gives 18 distinct shapes spread over 14 kcal/mol in MMFF energy. The two eosin Y conformers have the benzoate ring twisted one way or the other, which is why they are exactly degenerate.

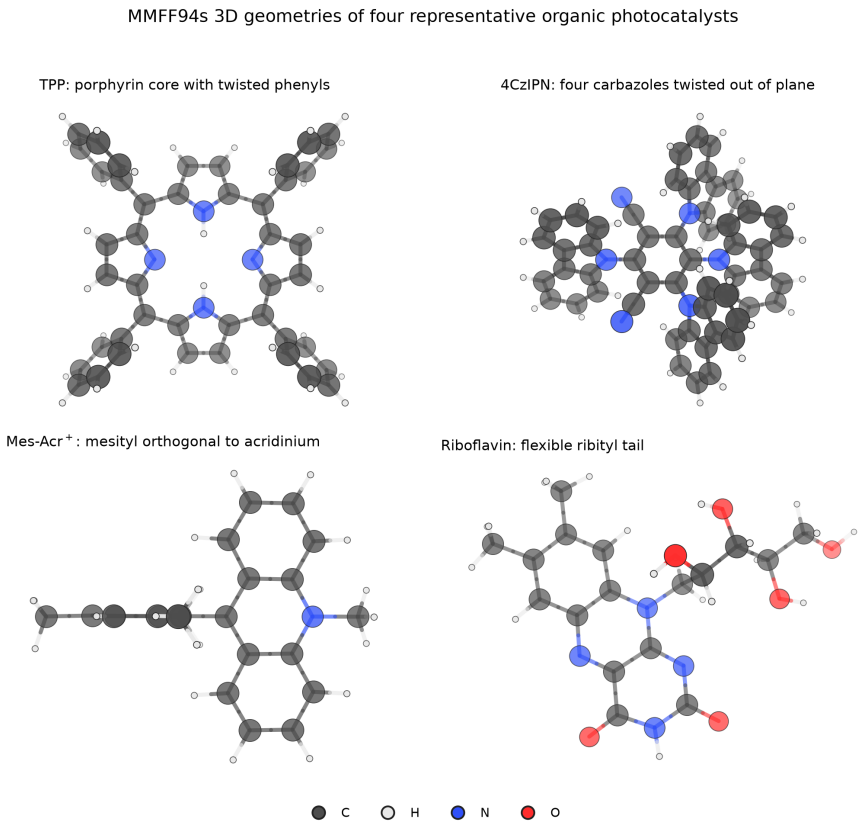


Figure 2: MMFF94s geometries. Each feature shown is a twist that decouples two π systems (TPP phenyls, 4CzIPN carbazoles, Mes-Acr$^+$ mesityl) or a flexible tail (riboflavin).

3 Step 2 — DFT geometry optimisation

3.1 What the script does

1. Builds the PySCF molecule from the MMFF xyz (`build_mol`). def2 effective core potentials are added automatically for elements heavier than Kr ($Z > 36$), i.e. iodine in rose bengal; bromine ($Z = 35$) does not need one.
2. Runs an SCF on the MMFF geometry, which gives the reference energy $E_{\text{MMFF geom}}$.
3. Optimises with **geomeTRIC**, which steps in internal coordinates (bonds, angles, torsions) rather than raw x, y, z.
4. Runs a final SCF at the optimised geometry, then compares the two structures with a **Kabsch-aligned RMSD**.

Concept to master

Kabsch alignment. Two copies of the same molecule can differ simply by being rotated or shifted. Kabsch's algorithm first centres both copies, then finds the rotation that best superimposes them (via a singular value decomposition of $P^{\mathsf{T}}Q$). Only then does it measure the per-atom distances. Without it, an RMSD would mix real structural change with an arbitrary orientation. The script reports RMSD over all atoms, over heavy atoms only, and the single largest atomic displacement.

3.2 Convergence criteria: `gau` vs `flat`

An optimisation stops when *all* of its tests pass. The default (`gau`, Gaussian-like) has three kinds of test:

Test	`gau`	`flat`
energy change per step	10^{-6} Ha	10^{-5} Ha
gradient (RMS / max)	3×10^{-4} / 4.5×10^{-4} Ha/bohr	same
displacement (RMS / max)	1.2×10^{-3} / 1.8×10^{-3} bohr	effectively off

For eosin Y, 27 `gau` steps met the gradient test, but the displacement test never settled. The optimiser kept taking sizeable steps along very soft torsions (the benzoate twist) at almost no cost in energy. The run was therefore resumed from its last frame with `--criteria flat`, which keeps the same gradient thresholds but drops the displacement test. It converged in 4 more steps. Riboflavin and rhodamine B used `flat` from the start.

Concept to master

Flat potential energy surfaces. The gradient tells you the surface is level; the displacement tells you the optimiser has stopped moving. On a very flat torsional surface the gradient can be small while the atoms still wander. Relaxing the displacement test is reasonable *because* the gradient test still holds. The structure is a true stationary point, but the exact torsion angle is poorly determined. Report it with that caveat.

3.3 Results

Molecule	AO	Cycles	ΔE (kcal/mol)	$\mathrm{RMSD}_{\mathrm{heavy}}$ (Å)	Max dev. (Å)	Criteria
thioxanthone	254	10	−1.88	0.030	0.06	gau
anthraquinone	264	15	−1.42	0.019	0.04	gau
benzophenone	246	11	−1.27	0.096	0.26	gau
riboflavin	478	18	−11.13	0.173	0.65	flat
eosin Y^{2-}	508	31	−17.73	0.316	1.30	gau → flat
rhodamine B^{+}	617	20	−20.23	0.398	1.27	flat

$\Delta E = E(\text{DFT-optimised}) - E(\text{MMFF geometry})$, both at the same DFT level. Wall times: rhodamine B 68 073 s (≈19 h), riboflavin 5 547 s (≈1.5 h); eosin Y's recorded 711 s covers only the 4-step resume.

Key finding

The earlier conclusion "MMFF is good enough, so skip DFT for the other 13" does not hold. It was based on three rigid molecules. The flexible and charged dyes moved roughly 6–13 times more than the rigid ones (heavy-atom RMSD 0.17–0.40 vs 0.02–0.03 Å) and dropped 11–20 kcal/mol in energy.

What actually moved. I aligned each MMFF and DFT pair and listed the atoms that moved most:

- **Eosin Y^{2-}**: the largest mover is a carboxylate oxygen (1.30 Å). The ring–ring twist changed from 98.6° to 86.1°.
- **Rhodamine B^{+}**: again a carboxylic-acid oxygen (1.27 Å), with the ring–ring twist almost unchanged (−84.9° → −88.9°). The next movers are ethyl hydrogens.
- **Riboflavin**: hydroxyl hydrogens and oxygens of the ribityl chain (up to 0.65 Å). This looks like a hydrogen-bond rearrangement.

So the large RMSDs come mainly from the **carboxyl groups and side chains rotating**, not from the chromophore cores. A likely reason is that MMFF minimised these ions without any solvent, while the DFT run used ddCOSMO water. A bare $-\mathrm{COO}-$ group in vacuum and a solvated one prefer different orientations. Because the HOMO and LUMO of these dyes sit on the cores, the orbital energies may change much less than the RMSD suggests. Step 5 exists to test exactly this (Section 6).

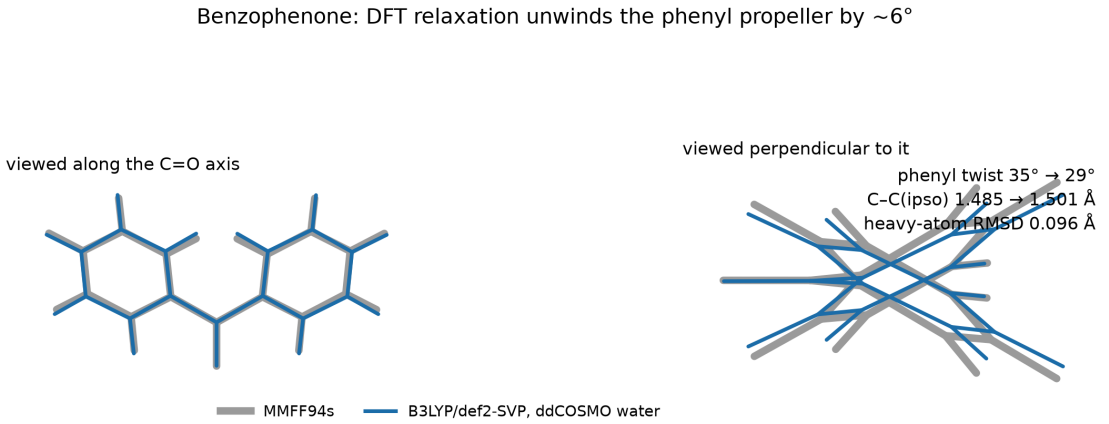
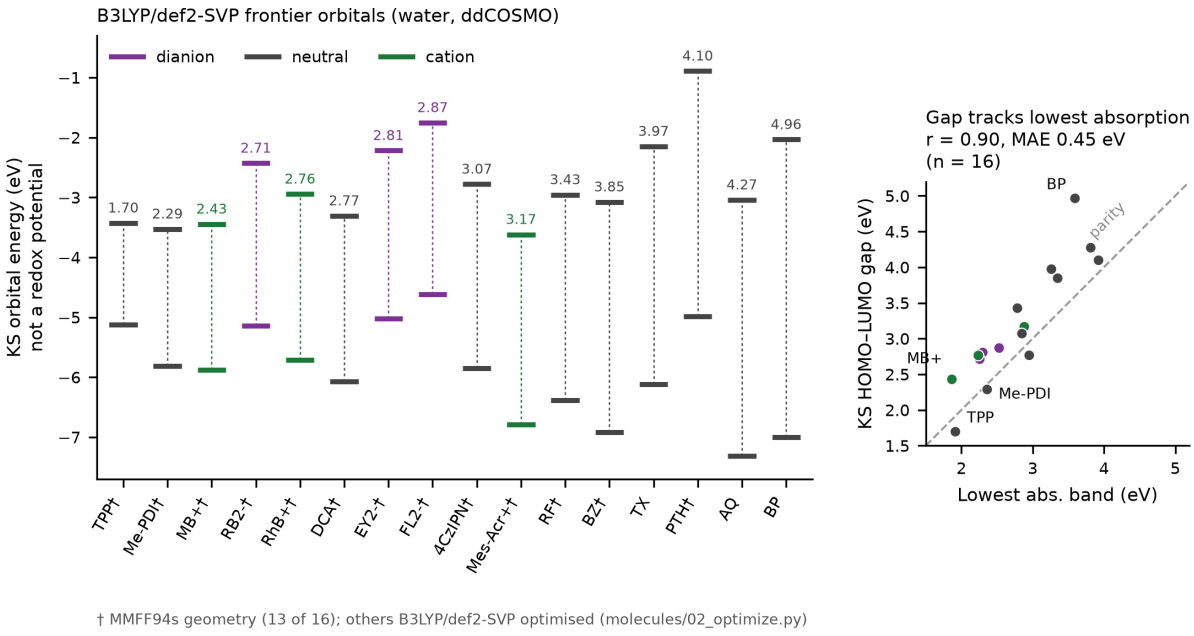


Figure 3: Benzophenone, MMFF94s (grey) vs DFT (blue): the change is invisible along C=O and obvious from the side. Always look from two directions.

4 Step 3 — Frontier orbital energies

A single point on each molecule's best available geometry: the DFT-optimised one if it exists, otherwise the MMFF one. The column `geom_source` records which was used. If the SCF fails to converge, the script retries with a second-order (Newton) solver. All 16 converged.



† MMFF94s geometry (13 of 16); others B3LYP/def2-SVP optimised (molecules/02_optimize.py)

Figure 4: Current diagram. Compared with the first version, the H^+/H_2 and O_2/H_2O lines were removed, the axis now says "not a redox potential", and † marks the 13 molecules still on MMFF geometry.

PC	HOMO (eV)	LUMO (eV)	Gap (eV)	Lowest abs. (eV)	Geometry
TPP	−5.12	−3.43	1.70	1.92	MMFF
Me-PDI	−5.81	−3.53	2.29	2.36	MMFF
MB^+	−5.87	−3.45	2.43	1.87	MMFF
RB^{2-}	−5.14	−2.43	2.71	2.26	MMFF
RhB^+	−5.71	−2.94	2.76	2.24	MMFF*
DCA	−6.07	−3.31	2.77	2.95	MMFF
EY^{2-}	−5.02	−2.22	2.81	2.30	MMFF*
FL^{2-}	−4.62	−1.75	2.87	2.53	MMFF
4CzIPN	−5.85	−2.78	3.07	2.85	MMFF
Mes-Acr^+	−6.79	−3.62	3.17	2.88	MMFF
RF	−6.38	−2.96	3.43	2.79	MMFF*
BZ	−6.92	−3.08	3.85	3.35	MMFF
TX	−6.12	−2.15	3.97	3.26	DFT
PTH	−4.99	−0.89	4.10	3.92	MMFF
AQ	−7.31	−3.04	4.27	3.82	DFT
BP	−7.00	−2.03	4.96	3.59	DFT

* a DFT geometry now exists, but step 3 has not been rerun on it.

Key finding

The Kohn–Sham gap tracks the lowest experimental absorption band ($r = 0.90$, MAE 0.45 eV, $n = 16$), and is larger than it for 13 of the 16. The largest outlier is benzophenone (+1.37 eV), whose weak lowest band is n→ π^*. TPP is compared against its $Q_x(0,0)$ band (647 nm), not the Soret band, which is S_2.

Concept to master

Three gaps that are not the same thing.

- *KS HOMO–LUMO gap*: a difference of orbital energies from one ground-state calculation.
- *Fundamental gap* = IE − EA: needs calculations on the cation and anion. For benzene, B3LYP's $-\varepsilon_{\mathrm{HOMO}}$ was 6.97 eV against an experimental IE of 9.24 eV.
- *Optical gap*: the lowest excitation, which includes the electron–hole attraction. It needs TDDFT.

A good correlation with absorption says the *trend* is captured, not that the absolute values mean anything. This is why the redox lines were removed from the diagram.

5 Step 4 — Where the orbitals live

For Mes-Acr^+ and eosin Y^{2-}, the script writes cube files for HOMO−2 to LUMO+1 (about 4.5 MB each, git-ignored). It then measures how much of each orbital sits on a "core" fragment, which is defined by a SMARTS pattern. The measure is the Mulliken population of orbital ϕ_k on the fragment's atoms:

$$p_k(\mathrm{core}) = \sum_{\mu \in \mathrm{core}} c_{\mu k}\,(\mathbf{S}\mathbf{c}_k)_\mu,$$

where $c_{\mu k}$ are the MO coefficients and $\mathbf{S}$ is the AO overlap matrix. Summed over all atoms this gives exactly 1.

Molecule	Orbital	E (eV)	Localisation
eosin Y^{2-}	HOMO	−5.02	91% xanthene core
	LUMO	−2.22	97% xanthene core
	HOMO−1, HOMO−2	−5.34, −5.83	88%, 84% benzoate
Mes-Acr^+	HOMO−2	−6.80	>99% mesityl
	HOMO−1	−6.79	96% mesityl
	HOMO	−6.79	99% acridinium
	LUMO	−3.62	98% acridinium

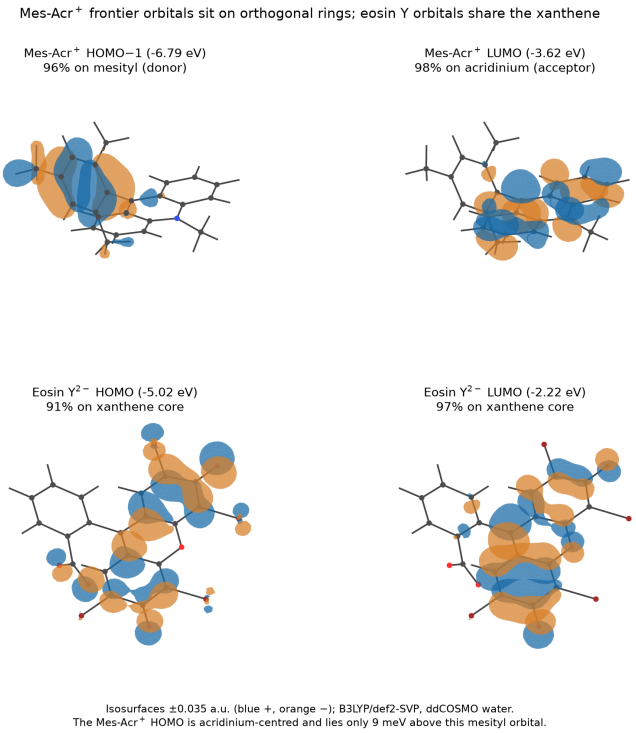


Figure 5: Isosurfaces at ±0.035 a.u., rendered by marching cubes in matplotlib (schematic: matplotlib only approximates which surface is in front).

Key finding

Eosin Y's lowest excitation is *local* ($\pi \to \pi^*$ on the xanthene). In Mes-Acr^+, donor (mesityl) and acceptor (acridinium) orbitals sit on two orthogonal rings, which gives *charge-transfer* character. This is the orbital-level picture of the 3D twist in Section 2.3. The top three occupied orbitals of Mes-Acr^+ lie within 13 meV of each other, so which one is labelled "HOMO" is not meaningful at this level of theory.

Caveat / open issue

Mulliken populations depend on the basis set. Use them for qualitative statements (91% vs 10%), not for small differences. The eosin Y orbitals in this step were computed on the MMFF geometry, before its DFT optimisation existed.

6 Step 5 — Does the geometry matter for the orbitals? (new, not yet run)

`05_geometry_check.py` answers the question raised by Section 3.3. For every molecule with a DFT geometry, it reruns the step-3 single point on the *MMFF* geometry and compares it with the step-

3 result on the *DFT* geometry. It writes the shifts ΔHOMO, ΔLUMO and Δgap (DFT minus MMFF) next to the RMSD and ΔE for each molecule.

Caveat / open issue

It cannot produce useful output yet. The script deliberately skips any molecule whose step-3 entry was not computed on the DFT geometry (`geom_source != "xyz_opt"`). That is currently true for eosin Y, riboflavin and rhodamine B: `frontier_orbitals.csv` was last written on 18 Oct at 14:33, before any of their optimisations finished. The order to run is:

```
python molecules/03_frontier.py --only eosinY riboflavin rhodamineB
python molecules/05_geometry_check.py
```

The three rigid molecules will then show the baseline (expected to be near zero). The three flexible ones will show whether rotating the carboxyl groups and side chains moves the frontier orbitals.

How to read the answer. If |Δgap| is small compared with the 0.45 eV MAE against experiment, MMFF geometries are adequate *for this property*, and the remaining 10 molecules need not be optimised. If it is comparable, every molecule needs DFT geometries before the orbital table can be trusted.

7 Infrastructure worth understanding

- **The dispersion workaround** (`utils/common.py`). On the Mac, `pyscf-dispersion` fails to load its compiled library. Because PySCF tries to import it, `import pyscf` itself crashed. The helper attempts the import, catches the `TypeError`, deletes the half-loaded PySCF modules from `sys.modules`, registers an empty placeholder module called `pyscf.dispersion`, and re-imports. PySCF then takes its own "no D3 available" path, and everything except dispersion works. This is why every script must import `utils.common` before `pyscf`, and why no result includes D3.
- **Density fitting** (`.density_fit()`) approximates the four-index electron repulsion integrals with an auxiliary basis. It is much faster at a very small accuracy cost, and is what made the 617-function rhodamine B optimisation feasible.
- **ddCOSMO** places the molecule in a cavity inside a dielectric continuum ($\varepsilon = 78.36$ for water). It matters most for the ions: a bare dianion in vacuum barely binds its extra electrons.
- **Provenance**: every xyz comment line records method, criteria, convergence and energy. Every CSV row records level of theory, geometry source and wall time.

8 Status board

Done	
Catalogue	16 organic photocatalysts, PubChem-checked, charge states documented
3D structures	Multi-conformer MMFF94s geometries for all 16; HTML viewer
DFT geometries	6 of 16 (3 rigid, 3 flexible/charged)
Orbitals	HOMO/LUMO for all 16; localisation for EY^{2-} and Mes-Acr^{+}
Figure fixes	Redox lines removed; MMFF-geometry molecules marked
Outstanding	
Rerun step 3	for EY, RF, RhB on their DFT geometries; then run step 5
Remaining geometries	10 molecules, depending on the step-5 result
TDDFT	not started; needed for any comparison with absorption energies
CT functional	CAM-B3LYP or ωB97X-D for Mes-Acr^{+} and 4CzIPN
Dispersion	D3 unavailable on the Mac; affects stacked or twisted structures
Solids (Week 2)	`solids/qe/` is empty; Materials Project and periodic DFT not started
Stale documents	
`README.md`, `REPORT.md`	last updated 17 Oct; don't mention the three new optimisations, the `flat` criteria, step 5, or the diagram changes
`data/catalysts.csv`	the original 29-entry list, superseded by `organic_photocatalysts.csv` but still present

9 Check yourself before meeting your PI

1. Why is eosin Y computed as EY^{2-}, and how does the InChIKey show it?
2. Why does eosin Y keep 2 conformers and riboflavin 18 after pruning?
3. What does Kabsch alignment remove before an RMSD is computed?
4. Why did eosin Y need the `flat` criteria, and why is that still a valid minimum?
5. The three flexible dyes have RMSDs of 0.17–0.40 Å. Which atoms moved, and why might that matter less for the orbitals than the RMSD suggests?
6. What exactly does step 5 compare, and why does it currently skip every flexible molecule?
7. Name three different "gaps" and the calculation each one needs.
8. Why is "the HOMO of Mes-Acr^{+}" not a meaningful label here?
9. What does the `pyscf.dispersion` workaround do, and what does it cost us?

10 Questions worth raising with your PI

- Is a geometry sensitivity below about 0.1 eV in the gap enough to keep MMFF geometries for the rest, or does the group want all molecules DFT-optimised regardless?
- For TDDFT, which functional does the group prefer (B3LYP for comparability, CAM-B3LYP/ωB97X-D for charge-transfer dyes)?
- Is there cluster time for the larger molecules? Rhodamine B alone took about 19 h.
- Should Week 2 (solids) start now in parallel, or after TDDFT?